

Lab: Building Your First Minimum Viable Prototype

Objective

Design and build (or plan in detail) a minimum viable prototype that tests the single most critical assumption of your invention idea. The goal is not a polished product but an informative action — a physical or experiential test that generates genuine prediction errors and updates your generative model. By the end of this lab, you will have identified your riskiest assumption, designed a test for it, and planned the perception-action loop that will drive your next iteration.

Materials and Prerequisites

- Your invention idea and causal model from previous labs
- Materials appropriate to your domain (these can be extremely simple: cardboard, tape, paper, string, clay, code, cooking ingredients, drawing tools, etc.)
- A timer or clock
- A partner or test user (optional but valuable)

Part 1: Identifying Your Riskiest Assumption (10 minutes)

Your generative model of your invention contains many assumptions — about how mechanisms work, what users want, how materials behave, and what the market will accept. Not all assumptions are equally risky. The riskiest assumption is the one that, if wrong, would make your entire invention fail.

List 5-7 key assumptions underlying your invention:

#	Assumption	If wrong, what happens?	How confident am I? (1-10)
1			

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Which assumption is the riskiest? (Highest impact if wrong AND lowest confidence)

Write your response here

Why is this the right assumption to test first?

Write your response here

Part 2: Designing the Minimum Viable Prototype (10 minutes)

Now design the simplest possible prototype that tests your riskiest assumption. The MVP should be: - As cheap and fast to build as possible - Focused on testing ONE specific thing - Designed so that both success AND failure are informative

What specific prediction are you testing? Write it as: "If I build [X], then [Y] will happen because [Z]."

Write your response here

Describe your MVP design: What will you build, draw, code, cook, or construct?

Write your response here

What materials do you need?

Write your response here

Estimated time to build:

Write your response here

What result would CONFIRM your assumption? Be specific.

Write your response here

What result would DISCONFIRM your assumption? Be specific.

Write your response here

If your assumption is disconfirmed, what will you do next? (This is important — plan for failure before you build.)

Write your response here

Part 3: Building or Detailed Planning (20 minutes)

If you have the materials and time, build your MVP now. If not, create a detailed build plan with sketches.

If Building:

Document your building process. Note moments where the material or system behaved differently than you expected.

What happened when you started building? Did anything surprise you?

Write your response here

Record of the perception-action loop during building:

Action I Took	What Happened (observation)	How I Adjusted

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If Planning:

Create a detailed build plan with sketches and step-by-step instructions.

Step-by-step build plan:

Write your response here

Sketch or describe the physical form of your prototype:

Write your response here

What materials will you acquire and when?

Write your response here

Part 4: Testing and Observation (10 minutes)

If you built the prototype, test it now. If you planned it, describe exactly how you would test it.

Test protocol: What will you do with the prototype and what will you measure or observe?

Write your response here

Results: What happened when you tested? (If planning, describe expected outcomes and what each would mean.)

Write your response here

Prediction error analysis:

Prediction	Actual Result	Prediction Error	What This Tells Me

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Write your response here

Was your riskiest assumption confirmed or disconfirmed?

Write your response here

How has your generative model of the invention changed based on this test?

Write your response here

Part 5: Planning the Next Iteration (10 minutes)

Based on what you learned, design the next step in your perception-action loop.

What is now the riskiest remaining assumption? (It may be the same one if the first test was inconclusive, or a different one if the first test resolved your primary uncertainty.)

Write your response here

What would the next prototype test?

Write your response here

How would the next prototype be different from this one?

Write your response here

Map your planned iteration trajectory (at least 3 steps):

Iteration	What It Tests	Type of Prototype	Expected Outcome
1 (this lab)			

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Write your response here

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Write your response here

|| 2 (next) |

Write your response here

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Write your response here

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Write your response here

|| 3 (future) |

Write your response here

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Write your response here

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Write your response here

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At what point would you shift from epistemic prototyping (learning) to pragmatic prototyping (demonstrating)?

Write your response here

Discussion and Debrief

1. **The building experience:** How did building (or detailed planning) change your understanding of your invention compared to thinking about it? What did you learn from materials that you could not have learned from mental simulation alone?
2. **Informative failure:** If your prototype failed to confirm your assumption, was the failure informative? Did you learn something specific, or was the failure ambiguous?
3. **Prototype scope:** Was your MVP truly minimal? Could you have tested the same assumption with an even simpler construction? What would you strip away?
4. **Perception-action speed:** How fast was your perception-action loop during building? What tools, materials, or environments would speed it up?
5. **The value of action:** Reflect on the difference between knowing something in theory and knowing it through building. What is the unique contribution of physical action to the inventive process?

Write your response here